

Introduction

To predict whether a patient would be readmitted within 30 days, we used the Diabetes 130-US hospitals for the years 1999–2008 dataset from the first dataset choice. Because it is linked to both the quality of patient treatment and the utilization of hospital resources, hospital readmission is a significant concern. The dataset contains 50 variables and 101,766 observations, including demographic data, hospital stay details, test results, prescription drugs, and diagnosis codes. We prepared and cleaned the data before generating the models. Variables with too many missing values were removed, and the remaining missing categorical values were replaced with “Unknown.” Additionally, we transformed the initial readmission variable into a binary outcome, coding patients readmitted within 30 days as "Yes" and all others as "No." This is an imbalanced classification challenge, as the positive class accounts for approximately 11.16% of the data.

To predict 30-day readmission, three methods are compared: logistic regression, classification tree, and random forest. We compared the models not only by accuracy but also by sensitivity, specificity, balanced accuracy, and AUC because the results are unbalanced. This project's primary objective is to identify the most effective model and the factors that are most crucial in predicting hospital readmission.

Data Preprocessing

Before model fitting, we cleaned the original hospital encounter dataset and prepared it for machine learning models. The main file, `diabetic_data.csv`, was used for modeling, while `IDS_mapping.csv` was used to interpret coded variables such as admission type, discharge disposition, and admission source.

Missing values were originally coded as "?" and were converted to NA. After checking missingness, we removed three variables with high missing rates: `weight`, `medical_specialty`, and `payer_code`. The variable `weight` had almost 97% missing values, while the other two also had substantial missingness. For the remaining categorical variables, missing values were replaced with "Unknown" so that patient records could be retained.

Variable	Missing Count	Missing Percent	Decision
weight	98,569	96.86%	Removed
medical_specialty	49,949	49.08%	Removed
payer_code	40,256	39.56%	Removed
race	2,273	2.23%	Replaced with Unknown
diag_3	1,423	1.40%	Grouped
diag_2	358	0.35%	Grouped
diag_1	21	0.02%	Grouped

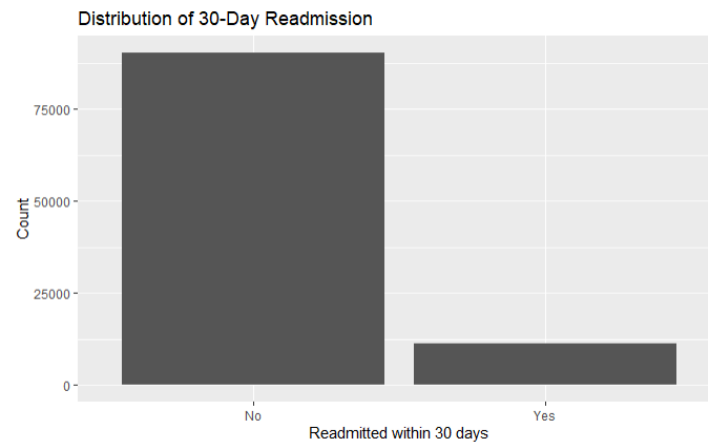
The original outcome variable, readmitted, had three categories: no readmission, readmission after more than 30 days, and readmission within 30 days. We converted it into a binary response variable, readmitted_30, where patients readmitted within 30 days were coded as "Yes" and all others as "No".

Identifier variables such as encounter_id and patient_nbr were removed because they do not provide meaningful predictive information. Coded categorical variables, including admission_type_id, discharge_disposition_id, and admission_source_id, were converted to factors. The diagnosis variables diag_1, diag_2, and diag_3 were grouped into broader clinical categories to reduce sparse levels while keeping useful diagnostic information.

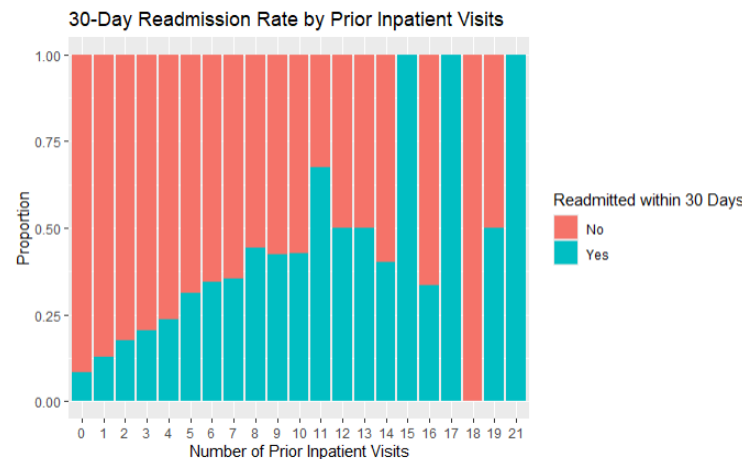
Exploratory Data Analysis

After preprocessing, we used EDA to examine the response distribution and key predictor patterns. The binary outcome was highly imbalanced: 90,409 encounters, or 88.84%, were not readmitted within 30 days, while 11,357 encounters, or 11.16%, were readmitted within 30 days. This suggests that accuracy alone may be misleading, so model evaluation should also consider

sensitivity, specificity, AUC, and the confusion matrix.



Prior inpatient visits showed the clearest relationship with early readmission. Patients with more previous inpatient visits had a higher proportion of 30-day readmission, suggesting that prior hospital utilization may be an important predictor of readmission risk.



Other variables showed weaker but still useful patterns. Age and primary diagnosis group varied slightly in readmission rates, while time in hospital was similar between the two groups. Overall, the EDA suggests a challenging classification problem because the outcome is imbalanced and individual predictors only moderately separate the two classes. However, prior healthcare utilization, demographics, diagnosis group, and hospital stay variables may still be useful when combined in machine learning models.

Modeling

Baseline Logistic Regression

After preprocessing and EDA, the cleaned data were split into training and testing sets using an 80/20 stratified split. Because the response variable was imbalanced, stratification was used to keep the outcome distribution similar in both sets. Both sets contained approximately 88.84% non-readmitted cases and 11.16% readmitted cases.

Before fitting logistic regression, factor variables with only one observed level in the training set were removed, including examide, citoglipton, and metformin.pioglitazone. These variables had no variation and could not provide useful predictive information. One sparse medication variable with an unseen level in the test set was also removed to avoid prediction errors.

Logistic regression was used as the baseline classification model because readmitted_30 is binary. The model estimates the probability of 30-day readmission and provides a simple, interpretable benchmark for comparison with more flexible machine learning models.

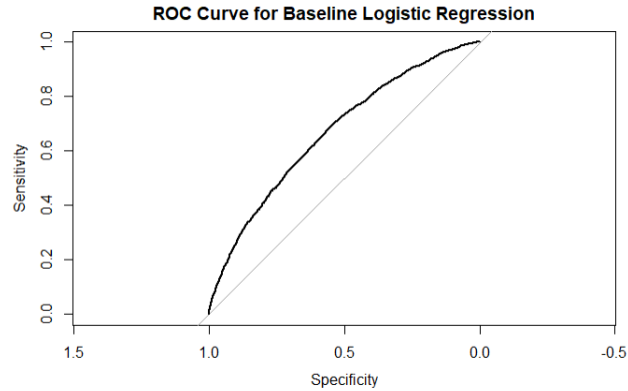
Using the default threshold of 0.5, the model achieved an accuracy of 88.84%. However, this was mainly due to the imbalanced outcome. The model identified only 39 of 2,271 truly readmitted cases, giving a very low sensitivity of 1.72%. This shows that the model was strongly biased toward predicting the majority class.

Lower thresholds were then examined. When the threshold was reduced to 0.2, sensitivity increased to 18.80%, while specificity decreased from 99.78% to 93.56%. This shows the trade-off between detecting more high-risk patients and creating more false positives.

Threshold <dbl>	Accuracy <dbl>	Sensitivity <dbl>	Specificity <dbl>	Balanced_Accuracy <dbl>	AUC <dbl>
0.5	0.8884139	0.01717305	0.9978430	0.5075080	0.6683012
0.3	0.8809454	0.07265522	0.9824678	0.5275615	0.6683012
0.2	0.8522013	0.18802290	0.9356230	0.5618230	0.6683012

3 rows

The AUC of the baseline logistic regression model was 0.6683. This suggests that the model has some ability to rank patients by readmission risk, but its overall discrimination is limited. Therefore, while logistic regression is useful as an interpretable baseline, it is not sufficient by itself for identifying high-risk readmission patients.



Overall, the logistic regression baseline shows that the prediction task is challenging due to class imbalance. The model provides moderate probability ranking ability based on AUC, but its classification performance is weak under the default threshold. These results justify comparing logistic regression with more flexible machine learning models in the next stage of the project.

Classification Tree

Next, we used a classification tree to estimate a patient's likelihood of being readmitted within 30 days. We used the identical processed training and test datasets as in the logistic regression analysis to guarantee consistency between approaches. Before modeling, variables with just one factor level and variables with levels not seen in the test set were eliminated. Confusion matrices, sensitivity, specificity, balanced accuracy, ROC curve, and AUC were computed at probability thresholds of 0.5, 0.3, and 0.2 to fit the classification tree on the training set and evaluate it on the test set.

The `number_inpatient` and `discharge_disposition_id` variables were significant predictors of 30-day readmission, as evidenced by the very simple classification tree (Figure 1). One benefit of the classification tree is that the initial splits provide an easy-to-understand way to determine which variables are most crucial and why the approach might work. The most significant role in the data in this instance is indicated by `number_inpatient`. However, the majority class ("No") was heavily favored by the tree. The model predicted nearly all observations as not being readmitted at thresholds of 0.5 and 0.3, yielding negligible sensitivity. The tree didn't start classifying some observations as "Yes" until the threshold was dropped to 0.2, at which point sensitivity increased to 0.2699. This demonstrates that while the model's overall capacity to

identify readmitted patients remained restricted, it could find certain positive cases only with a more lenient classification criterion.

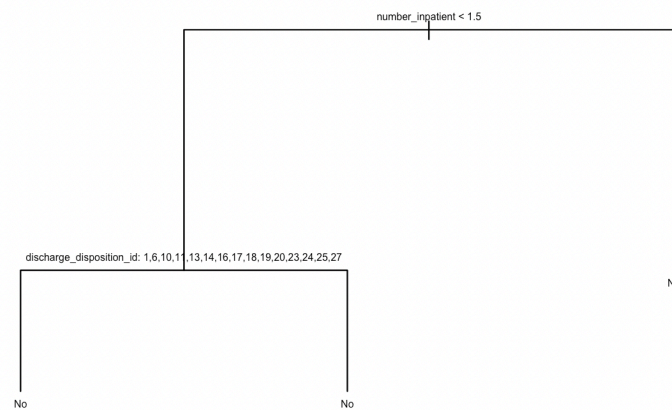


Figure 1. Classification tree for 30-day readmission. *Number_inpatient and discharge_disposition_id are the primary splitting points of the tree. The model exhibits a significant preference to classify data into the majority class because all terminal nodes are labeled "No."*

These findings demonstrate that although the classification tree is simple to understand, it performs poorly on this unbalanced dataset because the model is overly conservative when it comes to recognizing readmitted patients. In general, the classification tree is not the best prediction model in this project, but rather a good baseline strategy.

Random Forest

Next, to predict a 30-day readmission, we used a random forest model. In contrast to a single classification tree, a random forest improves stability and predictive accuracy by building a large number of trees using bootstrap samples and combining their predictions. When a single tree is too basic or unstable, this technique is very helpful. Using confusion matrices, sensitivity, specificity, balanced accuracy, ROC curve, and AUC under probability thresholds of 0.5, 0.3, and 0.2, we evaluated the random forest model on the test set.

The random forest model outperformed the single classification tree and produced a more balanced classification outcome. Random forest outperformed the tree model in terms of sensitivity and balanced accuracy, particularly when the threshold was lowered to 0.2. According to our findings, the model's sensitivity improved to roughly 0.2721 while retaining respectable specificity, and it achieved the maximum balanced accuracy at a threshold of 0.2. With an AUC of roughly 0.6569, the ROC curve likewise demonstrated superior discrimination compared to

the classification tree (Figure 3).

Discharge_disposition_id, number_inpatient, num_lab_procedures, num_medications, time_in_hospital, and diagnosis group variables were among the most significant predictors, according to the variable importance plot (Figure 2). This implies that 30-day readmission prediction is significantly influenced by both past hospital use and clinical traits.

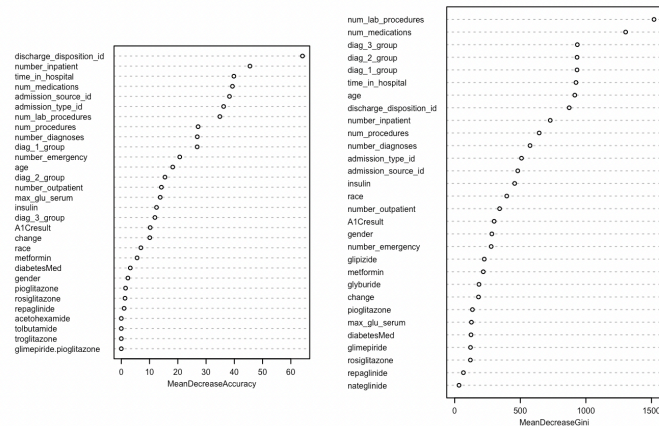


Figure 2. Variable importance plot for the random forest model. The plot shows the most important predictors for 30-day readmission. Variables such as *discharge_disposition_id*, *number_inpatient*, *num_lab_procedures*, *num_medications*, and diagnosis group variables were among the most influential predictors.

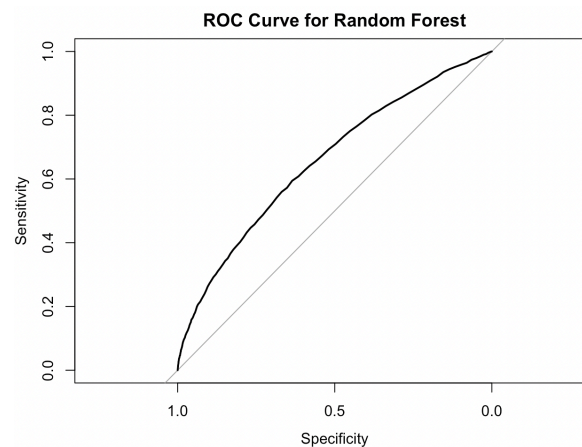


Figure 3. ROC curve for the random forest model. The random forest model's classification performance across various probability thresholds is summed up by the ROC curve. The model outperforms random guessing in differentiating between readmitted and non-readmitted patients, as seen by the curve's location above the diagonal reference line.

Overall, by identifying more positive readmission cases and achieving a better balance between sensitivity and specificity, Random Forest addressed the primary shortcoming of the classification tree. The model is more stable than a single tree since it incorporates several trees.

All Three Models Comparison

Table 1. Summary comparison of the three models.

Model	AUC	Balanced Accuracy (0.2)	Sensitivity (0.2)	Specificity (0.2)
Logistic Regression	0.6683	0.5618	0.1880	0.9356
Classification Tree	0.5968	0.5708	0.2699	0.8717
Random Forest	0.6569	0.5855	0.2721	0.8988

In Table 1, the three models had varying strengths. The best overall ability to differentiate between patients who were readmitted and those who weren't was demonstrated by logistic regression, which had the highest AUC (0.6683). Nevertheless, many patients who were really readmitted were still neglected because its sensitivity at threshold 0.2 was just 0.1880. The classification tree performed the worst among the three models since it had the lowest AUC (0.5968). Its specificity decreased to 0.8717 at threshold 0.2, despite an increase in sensitivity to 0.2699. This implies that while the single tree was still helpful in identifying the significant variables, it was not as stable as the other two approaches. Random forest provided the best balance. It exhibited the highest balanced accuracy (0.5855) and sensitivity (0.2721) at threshold 0.2, but its AUC (0.6569) was marginally lower than logistic regression. It maintained a comparatively high specificity (0.8988). Because it provided an optimal compromise between identifying positive readmission cases and preserving overall classification performance, the random forest was the most appropriate model for our project.